

Nicholas Lasagna

Berkeley, CA | nicholaslasagna@gmail.com | [linkedin.com/in/nicholas-lasagna-798118277](https://www.linkedin.com/in/nicholas-lasagna-798118277) | github.com/nicholaslasagna | nicholaslasagna.com

EDUCATION

Texas Tech University

Bachelor of Science in Computer Science, Minor: Mathematics | GPA: 3.565 / 4.00

Lubbock, TX

Expected May 2027

Accelerated M.S. Computer Science | GPA: 4.00 / 4.00

Expected 2028

Accelerated Program: Enrolled in an accelerated BS/MS Computer Science program, completing graduate-level coursework during an undergraduate degree.

Relevant Coursework: Data Structures & Algorithms, Analysis of Algorithms, Computer Architecture, Systems & Assembly, Theory of Automata, Software Engineering, Engineering Statistics, Differential Equations.

TECHNICAL SKILLS

Languages: Java, Python, Rust, C++, C, C#, TypeScript, JavaScript, SQL, x64 Assembly, HTML/CSS

Infrastructure & Cloud: Linux (Ubuntu), Oracle Cloud Infrastructure (OCI), Docker, Git, REST APIs, CI/CD

Systems & Backend: Distributed Systems, PostgreSQL, Supabase, JVM Tuning, Runtime Debugging, System Monitoring, LLMs

Engines & Frameworks: Unreal Engine 4/5, Unity, Clickteam Fusion 2.5, Next.js, Blender

EXPERIENCE

Imagicast Studios | Berkeley, CA | Co-Founder & Lead Developer

2021–Present

Independent software and game development

- Own core systems architecture for an asymmetrical multiplayer game in active development, from initial design through ongoing multiplayer implementation.
- Engineer engine-level gameplay systems in C++ and C# on Unreal Engine 4/5: state management, input handling, and runtime multiplayer logic.
- Designed and implemented distinctive multiplayer roles within a 6v1v1 gameplay loop.
- Coordinate distributed development workflows on Git with iterative, agile practices across evolving requirements.

SYSTEMS & ENGINEERING PROJECTS

RealFiction Distributed Multiplayer Server Network (Java, Oracle Cloud (OCI), Ubuntu Linux, Distributed Systems)

- Architected, deployed, and operate a live multi-server Minecraft network on OCI, serving concurrent real-time player workloads in production.
- Own production operations end to end: availability and latency monitoring under real player load, process monitoring, and live debugging of running services.
- Built and maintain custom Java backend services and persistent player data systems across multiple servers.
- Configured networking, permissions, and deployment pipelines; performed JVM tuning and runtime optimization.

RealLang (Compiler Engineering, Structured Diagnostics)

- Designed and implemented a working programming language whose diagnostics are structured and machine-applicable, targeting LLM code generation and repair.
- Built the full front-end pipeline (lexer, parser, type checker, diagnostic emitter), with error output consumable by both humans and AI tooling.

RealForge Workbench (Local-First AI Tooling, TypeScript)

- Building a desktop workbench that connects to a user-configured, OpenAI-compatible local model, opens a project, and proposes code changes for human review before apply.
- Designed the safety architecture: bounded context, untrusted-by-default output labeling, dry-run previews, approval gates, and automated privacy scans before any change lands.
- Bootstrapped the workbench's initial foundation on RealLang.

HeartHaven (Next.js, TypeScript, Supabase, PostgreSQL)

- Developing a browser-based multiplayer platform with live sessions, room customization, social interaction, and persistent world state.
- Implementing authentication, persistent player state, real-time synchronization, and version checks that detect conflicting concurrent updates.
- Designing a PostgreSQL schema with Row-Level Security (RLS) for progression, inventories, and live user data.

Runtime Tooling & Low-Level Systems Project (Rust, Memory Safety, Concurrency)

- Engineered performance-sensitive Rust runtime tooling for a game modification project with an active community.
- Debugged complex runtime behavior involving concurrent execution and systems-level resource management, prioritizing memory safety and correctness for stability under load.

LEADERSHIP & INVOLVEMENT

National Society of Leadership and Success: Active Member.